Roll No. ....

TEE–701

B. TECH. (EE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ELECTRICAL DRIVES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are advantages of electric drives ? Discuss. (CO1)

OR

(b) State essential component of electrical drives. (CO1)

2. (a) What are the main factors which decides the choice of electrical drive for a given application ? (CO3)

OR

(b) A horizontal conveyer belt moving at uniform speed of 1.2 m/s transport material at the rate of 100 tones per hour belt is 200 m long and driven by motor at 1200 r.p.m. Determine the load inertia referred to motor shaft. (CO3)

3. (a) Calculate starting time of drive with the following parameters J = 10 km-m^2 T = 15 + 0.5 W_m and TI = 5 + 0.6 W_m. (CO2)

P. T. O.

OR

(b) Derive fundamental torque equation in term of dynamic torque. (CO2)

4. (a) Discuss four quadrant operation of electrical drive using hoist load. (CO3)

OR

(b) Discuss steady state stability limit. How is it used to decide stability of the system ? (CO3)

5. (a) Discuss load equalization. (CO2)

OR

(b) A drive has the following parameters: $J = 10$ km-m^2 $T = 100-0.1$ N and TI = 0.05 N, where N is speed in r.p.m. initially the drive is operating in steady state now it is to be reversed for this motor characteristic is changed to $T = -100-0.1$ N calculate the time of reversal. (CO2)

Roll No. ....

TEE–702

B. TECH. (EE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

NON-CONVENTIONAL RESOURCES AND ENERGY SYSTEM

(NCR & ES)

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you mean by Non-conventional Resources ? Explain in detail. (CO1)

OR

(b) Give the detailed description of Energy System. (CO1)

2. (a) Give the classification of Energy sources. (CO1), (CO2)

OR

(b) What are the characteristics of solar cell. (CO1), (CO2)

3. (a) What are the salient features of Non-Conventional Energy ? (CO1), (CO2)

OR

(b) Explain the types of solar collectors. (CO1), (CO2)

P. T. O.

4. (a) Give the comparison of fossil fuel-based energy resources and alternate energy resources. (CO1)

OR

(b) "Economy and Social development is related to Renewable energy", give explanation in evidence to this. (CO1)

5. (a) Discuss the various environmental aspects of energy. (CO1), (CO2)

OR

(b) Explain in brief : (CO1), (CO2)

(i) Solar thermal system

(ii) Extraterrestrial radiation

Roll No. ....

TOE–703

B. TECH. (EE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

BIOMEDICAL ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Provide a brief overview of the ECG waveform, electrodes and common uses of ECG in medical diagnosis and monitoring. (CO1)

OR

(b) Describe the concepts of action potential and resting potential in neurons with appropriate waveforms. (CO1)

2. (a) What are the problems associated with measurement of a living system. (CO1)

OR

(b) Explain the principles of Electroencephalography (EEG) and describe the main types of brain waves typically observed in EEG recordings. (CO1)

P. T. O.

3. (a) Explain a man instrument system with the help of a block diagram. (CO2)

OR

(b) Discuss the main objectives of main instrument system which are to be fulfilled while designing. (CO2)

4. (a) Explain the importance of transducers in measurement of a living system. (CO2)

OR

(b) Define the term 'Biometrics.' List any five factors to be considered for the design and specifications of medical instrumentation systems. (CO2)

5. (a) Explain the respiratory system of human body. (CO2)

OR

(b) Discuss in brief the biopotential electrodes and its types. (CO2)

Roll No. ....

TOE–704

B. TECH. (ELECTRICAL ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

FUNDAMENTAL OF IoT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the key characteristics (at least five) of IoT technology using suitable examples. (CO1)

OR

(b) Discuss any *two* major challenges in Internet of things technology and justify your answer with building suitable scenario, impacts, and their mitigation method. (CO1)

2. (a) Western Rajasthan is arid region and suffers from scarcity of water where water storage and management is a major issue. Consider yourself an IoT engineer, design and explain key components which you will suggest for building a IoT enabled water storage supply systems for arid regions. (CO1)

P. T. O.

OR

(b) Explain any *three* IoT terminologies given below : (CO1)

(i) Layers

(ii) Node

(iii) Firmware

(iv) Actuators

3. (a) What do you mean by interoperability ? Explain its benefit in IoT systems. (CO2)

OR

(b) Dehradun city is moving forward in its smart city project. Being and IoT engineer, what would be recommendation for selection of resources and their potential features ? (CO2)

4. (a) Consider a scenario where a manufacturing company wants to optimize its production processes using IoT technology. Explain which key IoT technologies and components they should consider integrating into their operations and how these technologies can improve efficiency and productivity. (CO2)

OR

(b) Explain data accumulation layer and data abstraction layer. What are the key differences between both layers ? (CO2)

5. (a) Explain the network topologies used in IoT technology with suitable diagrams and application. (CO2)

OR

(b) Discuss various methods for object identification in IoT technology and also explain the working of RFID. (CO2)

Roll No. ....

TEE–706

B. TECH. (EE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

UTILIZATION OF ELECTRIC ENERGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the advantages of electrically produced heat ? What are the properties to be possessed by the element used in resistance oven ? (CO1)

OR

(b) Differentiate between direct and indirect type resistance heating. (CO1)

2. (a) List out the advantages and explain about the applications of dielectric heating. (CO1)

OR

(b) Dielectric heating is to be employed to heat a slab of insulating material 20 mm thick and 1530 mm^2 in area. Power required is 200 W and a frequency of 3 MHz is to be used. The material has a permitivity of 5 and p.f. of 0.05. Determine the voltage necessary and the current which will flow through the material. (CO1)

P. T. O.

3. (a) A 20-kW, 1-phase, 220-V resistance oven employs circular nickel chrome wire for its heating elements. If the wire temperature is not to exceed 1107° C and the temperature of the charge is to be 500°C, calculate the length and size of wire required. Assume a radiating efficiency of 0.6 and the specific resistance of nickel chrome as $100 \times 10^{-6}\,\Omega$ cm and emissivity 0.9. (CO1)

OR

(b) What are different methods of heat transfer ? Explain in brief. (CO1)

4. (a) Discuss the Laws of Illumination. (CO2)

OR

(b) A room 6 m × 4 m is illuminated by a single lamp of 100 C.P. in all directions suspended at the centre 3 m above the floor level. Calculate the illumination (i) below the lamp and (ii) at the corner of the room. (CO2)

5. (a) Define the terms : (i) Lux (ii) Luminous Flux (iii) Candle Power. (CO2)

OR

(b) A school classroom, 7 m × 10 m × 4 m high is to be illuminated to 135 lm/m² on the working plane. If the coefficient of utilization is 0.45 and the sources give 13 lumens per watt, work out the total wattage required, assuming a depreciation factor of 0.8. Sketch roughly the plan of the room, showing suitable positions for fittings, giving reasons for the positions chosen. (CO2)